Course Description: Academic Year 2026-27

Course Title and Code: Applied Physics (AS1108)	
Teaching Scheme and no. of hours per week	L-T-P: 3-0-0 (3 hours)
Credits	3
Programme(s) and Batch	B. Tech.
Semester (as per curriculum structure)	I Sem
NCrF Level (4.5/5.0/5.5/6.0)	4.5
Pre-requisite course (Course Code and Title)	Nil
Desired prior knowledge/skills	-
Weightage	Theory 100%
The occupational path it may lead to, if any	-

Course Objective: This physics course provides a comprehensive exploration of classical mechanics, electromagnetism, and quantum mechanics, equipping students with a deep understanding of fundamental physical principles and their real-world applications.

Learning Outcomes:

On successful completion of this course, the students should be able to:

AS1108.1 Explain and apply conservation principles, including the conservation of energy and momentum in rectilinear, polar, and spherical coordinates.

AS1108.2 Calculate and analyze work, energy, and momentum in various physical systems.

AS1108.3 Understand Eddy currents and Maxwell's equations in the context of electromagnetism.

AS1108.4 Understand the wave-particle duality and its significance in quantum physics.

AS1108.5 Solve the one-dimensional Schrödinger equation for various physical systems.

Syllabus

Section – I Classical Mechanics

Review of basic concepts, Rectilinear, polar, and spherical coordinates; Motion in 1-, 2-, and 3-dimensions; Conservation principles; Dynamics of particles in linear and circular motion, Work and energy, Momentum and impulse, Momentum conservation and collisions, Centre of mass, Moment of inertia.

Section – II Electromagnetism

Electrostatics: Gauss's law, Electrostatic potential, Conductors, capacitors, and dielectrics, Electromotive force, Energy and power in electric circuits.

Magnetostatics: Lorentz force, Motion of charged particles and applications, Force and torque on a current loop, DC motor, Hall effect, Ampere's laws, Magnetic materials, and magnetization.

Electrodynamics: Faraday's law, Lenz's law, Motional electromotive force, Induced electric fields, Eddy current, Maxwell's equation, Wave solution of Maxwell equation.

Section – III Quantum Mechanics

Introduction, Photoelectric effect, Compton scattering and pair production, Wave-particle duality, probability and uncertainty, De Broglie's hypothesis, Wave function and wave packets, 1-dimensional Schrodinger equation, Particle in a box.

Textbooks/ References:

1. Cutnell, J. D., Johnson, K. W., Young, D., & Stadler, S. (2016). Physics. John Wiley & Sons.
2. Halliday, D., Resnick, R. and Walker, J. (2011) Fundamentals of physics. 9th Edition, John Wiley & Sons Inc.
3. Young, H.D. and Freedman, R.A. (2004) Sears and Zemansky's University Physics with Modern Physics. 10th Edition, Pearson Addison Wesley.
4. D. Kleppner and R. J. Kolenkow (2017) An Introduction to Mechanics. 1st Edition, McGraw Hill Education.
5. D. J. Griffiths (2020) Introduction to Electrodynamics, 4th Edition, Cambridge University Press.
6. Griffiths, D. J., & Schroeter, D. F. (2018). *Introduction to quantum mechanics*. Cambridge university press.

Online resources:

1. The Feynman Lectures on Physics, FLP Vol. I Table of Contents (caltech.edu)
2. Electromagnetics And Applications, MIT, Electromagnetics and Applications | Electrical Engineering and Computer Science | MIT OpenCourseWare
3. Quantum Physics I, MIT, Quantum Physics I | Physics | MIT OpenCourseWare
4. Lecture Series on Quantum Physics by Prof.V.Balakrishnan, Department of Physics, IIT Madras, NPTEL

Following are the evaluation schemes for each Institute as per Gradebook

Institute of Engineering and Technology

Sr. No	Specifications	Weightage	Marks
1	Class Participation	Nil	Nil
2	Assignment	15	15
3	Quiz	15	15
4	Mid-Term Examination (Theory)	20	20
5	End–Term Examination (Theory)	40	40
6	Report	Nil	Nil
7	Project	10	10
8	Lab Evaluation	Nil	Nil
9	Course Portfolio	Nil	Nil
10	Presentation	Nil	Nil
11	Viva	Nil	Nil
	Total	**100**	**100**

Attendance Policy

Every student is expected to put in 100% attendance. However, the student with attendance below 75% in a course, shall be penalized as follows.

- 70.0% to less than 75.0% attendance: A warning letter will be issued.
- 50.0% to less than 70.0% attendance: One grade drop. However, a D grade will remain D.
- Less than 50% attendance: Two grades drop. However, a D grade will remain D.

Note:

1. *This Relaxation in attendance includes participation in all kinds of activities, medical emergencies, participation in placement opportunities, late registrations, disciplinary actions, or any other reasons not mentioned here.*